RESERVOIR ENGINEERING — EXAMPLE STUDY

Simulation Study: History Match & 10-Year Forecast

The **BLACK_OIL_DEMO** field model was history matched and forecast end-to-end by an AI agent connected to **tNavigator** through the Local MCP Server. The engineer supervised each step and made the final decisions; the agent ran the models and the diagnostics, and produced this report.

616	65.1 → 2.77	54% → 7.3%	12% → 1.6%	2.46 M sm^3
SIMULATIONS RUN	OBJECTIVE FUNCTION	FIELD WATER ERROR	FIELD OIL ERROR	FIELD EUR

Scope of work

Differential-evolution history matching with a 200-run sensitivity screening, per-well segmented decline-curve analysis, and a 10-year production forecast cross-checked against the matched simulation model.

Dataset & tools

tNavigator BLACK_OIL_DEMO tutorial model (synthetic); history 2011-05-15 to 2015-02-01; 15 producers. tNavigator 26.2 with the Local MCP Server; the AI agent drives the full workflow API.

“The AI agent does the work — the engineer stays the engineer.”

Study workflow

1. **Connect** — the AI agent attaches to tNavigator through the Local MCP Server, one installer component launched from the GUI.
2. **Execute** — decks, 616 simulation runs, sensitivity screening and DCA, driven through the tNavigator workflow API.
3. **Supervise** — the engineer reviews diagnostics at every step, redirects methodology, and signs off the match and forecast.
4. **Report** — the agent generates interactive dashboards and this document automatically.

About this example. Prepared by Rock Flow Dynamics with tNavigator 26.2 to demonstrate AI-agent reporting in a TIMOR GAP-style template. All results derive from the synthetic BLACK_OIL_DEMO tutorial dataset; this document is not an official TIMOR GAP publication and contains no TIMOR GAP data.

SECTION 01

Assisted History Matching

The agent began with rate matching, which produced a convincing first result — until measured bottom-hole pressure exposed it as a false optimum. Skins, permeability corridors and completion hypotheses were then evaluated individually, a 200-run sensitivity study identified the influential parameters, and differential evolution converged on the final match with BHP included in the objective.

Indicator	Before	After
Objective function	65.1	**2.77**
Cumulative field water error	54%	**7.3%**
Cumulative field oil error	12%	**1.6%**
BHP gauge RMSE (W3 / W40)	—	**23% / 13%**
Selected match	—	**v754**

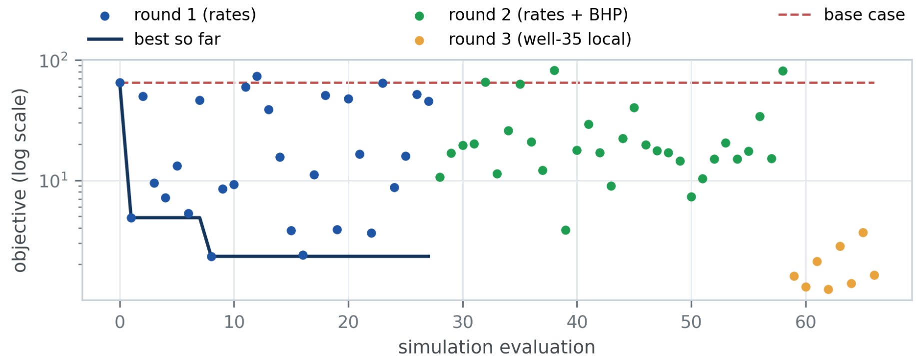

Figure 1. Optimisation convergence: round 1 matched rates only; round 2 added measured BHP to the objective; round 3 refined well 35 locally. The dashed line is the base case.

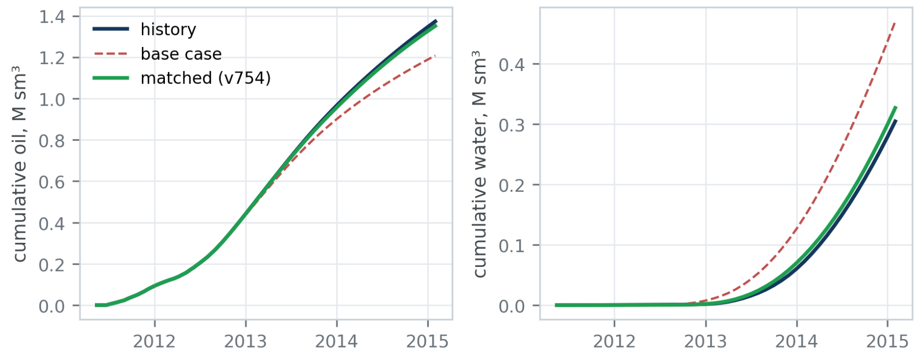

Figure 2. Cumulative field oil and water: history, base case and the selected match (v754).

Engineering control. The agent proposed each step; the engineer rejected an artificial skin adjustment, demanded the sensitivity-first methodology and made the final Pareto trade-off between rate and pressure objectives. Every iteration loop took minutes, not days.

SECTION 02

10-Year Production Forecast

Per-well decline-curve analysis was performed on the matched model, with one Arps fit per decline cycle and AICc-based model selection across harmonic, exponential and hyperbolic forms. Nine of fifteen wells qualified for forecasting; the remainder had converted to injection. The DCA forecast was cross-checked against the simulation forecast as a physics feasibility check.

Forecast indicator	Value
Wells forecast	**9 / 15**
DCA 10-year volume	**1.31 M sm^3**
Simulated 10-year volume	**1.09 M sm^3**
DCA vs simulation gap	**17%**
Field EUR (produced + forecast)	**2.46 M sm^3**

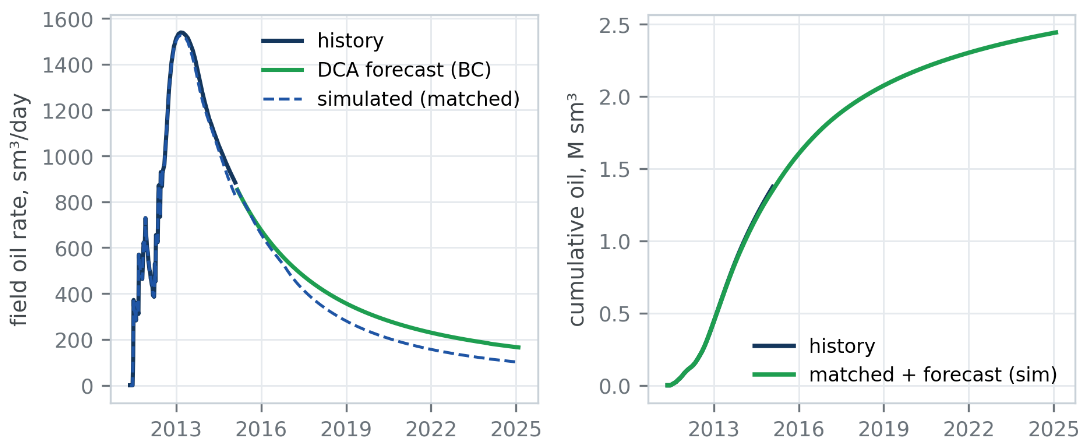

Figure 3. Field oil rate and cumulative production: history, DCA forecast under boundary conditions, and the matched-model simulation forecast used as the feasibility cross-check.

Interpretation. The 17% gap between decline-curve and simulation forecasts is driven by pressure support in the matched model that empirical declines cannot see — a reminder to anchor empirical forecasts to physics before reserves are booked.

Explore the live results

The interactive dashboards behind Figures 1–3 are published online:

- History match: `lanetszb.github.io/tnav-ai-webinar/dashboards/hm_dashboard.html`
- DCA forecast: `lanetszb.github.io/tnav-ai-webinar/dashboards/dca_dashboard.html`

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